

# Pooja Kiran Bharadwaj

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## EDUCATION

### Arizona State University

Tempe, AZ

*Master of Science in Information Technology*

Aug. 2024 – May 2026

- Coursework: Machine Learning Security, Cloud Security Architecture, Information Assurance, Network Defense, Secure Systems

### M. S. Ramaiah University of Applied Sciences

Bengaluru, India

*Bachelor of Technology in Computer Science and Engineering*

Aug. 2019 – Aug. 2023

## EXPERIENCE

### AI Security Researcher & Engineer

Aug. 2024 – Present

*Independent Security Research (Concurrent with M.S. program)*

Tempe, AZ

- Engineered an inline MCP security gateway with a 5-layer decision pipeline and 50+ prompt-injection detection rules by implementing JSON-RPC parsing, Unicode/homoglyph normalization, and real-time policy enforcement in Python/FastAPI, achieving 0.015ms average inspection latency per tool call across 50,000 benchmarked iterations
- Developed a static IAM policy analyzer for AI agent roles on AWS by implementing 25 deterministic rules covering wildcard permissions, privilege escalation, audit-trail tampering, and credential-harvest chains with SARIF output, enabling automated least-privilege enforcement as a CI merge gate with zero runtime dependencies
- Built a model supply-chain scanner with a taint engine and symbolic resolver covering 17 file formats by implementing deep opcode analysis for pickle Protocol 0–5, SafeTensors header inspection, GGUF structural parsing, and typosquat detection, achieving 100% detection on 12 attack fixtures (2 CVEs) in 116ms with 0% false positive rate
- Created adversarial prompt-generation and model-extraction attack harnesses (FGSM, PGD, C&W) mapped to MITRE ATLAS by implementing multi-category red-team generation and detector evaluation with structured SARIF evidence output, producing reproducible security assessments for guardrail validation and input/output filtering decisions

### Business Compliance Lead & Market/Cost Analyst

Aug. 2025 – Dec. 2025

*AEROSEC – ASU Technology Innovation Lab (Honeywell Aerospace)*

Tempe, AZ

- Led regulatory compliance analysis for Honeywell Aerospace's security product by mapping requirements and conducting competitive cost modeling across aerospace frameworks, delivering a compliance framework supporting the go-to-market strategy

### Graduate Teaching Assistant / IT Grader

Jan. 2025 – Oct. 2025

*Ira A. Fulton Schools of Engineering, Arizona State University*

Tempe, AZ

- Evaluated technical assignments for 100+ students per semester by assessing cloud security architectures, network defense configurations, and IAM access control implementations, maintaining consistent grading standards across multiple course sections

## PROJECTS

### MCP Agent Security Gateway | *Python, FastAPI, Docker, Kubernetes, GitHub Actions*

- Architected an inline stdio proxy intercepting MCP tools/call JSON-RPC requests through a 5-layer pipeline (server trust, tool-call policy, process-spawn evaluation, semantic intent, network egress) by implementing allow/block/redact/quarantine decisions, preventing unauthorized tool execution before requests reach downstream servers
- Implemented 50+ prompt-injection detection rules with Unicode and homoglyph normalization by building PII detection for 9 data types and exfiltration-pattern matching, achieving comprehensive coverage across instruction override, role manipulation, delimiter injection, and encoded payload attacks
- Engineered production-grade operational controls by integrating SHA-256 hash-chained audit logging, circuit breakers, rate limiting, shadow mode, and write-ahead logging with Kubernetes deployment manifests, providing tamper-evident observability with full traceability across all security decisions

- Validated security controls with 529 automated tests at 77% coverage across Python 3.10/3.11/3.12 by integrating Bandit, CodeQL, Trivy, Gype, and pip-audit into the CI pipeline, achieving p50 latency of 0.0123ms and p99 of 0.04ms with near-zero performance overhead

#### **AWS Agent Identity Guard** | *Python, AWS IAM, Terraform, SARIF*

- Developed 25 deterministic IAM rules targeting AI agent roles by detecting wildcard service prefixes, iam:PassRole without conditions, privilege-management actions, audit-trail tampering, and credential-harvest plus lateral-movement chains, covering the full agent identity attack surface for Bedrock, SageMaker, and Lambda workloads
- Implemented credential-harvest chain detection by analyzing combined permission patterns across sts:AssumeRole, cross-account trust without ExternalId, and network egress modifications, identifying multi-step privilege escalation paths in a single policy scan
- Built a CI merge gate by outputting findings in Text, JSON, and SARIF formats with exit code 1 on critical/high severity, integrating with GitHub Code Scanning for inline PR annotations that block deployment of over-permissioned agent roles
- Achieved zero runtime dependencies for local policy scanning by implementing pure-Python static analysis with optional boto3 live-account mode, Terraform infrastructure examples, and documented threat model with security audit

#### **HF Model Provenance Scanner** | *Python, Hugging Face API, Docker, SARIF*

- Engineered a taint engine and symbolic resolver for pickle opcode analysis (Protocol 0–5) by implementing deep REDUCE/GLOBAL/STACK\_GLOBAL walks with gadget-chain matching, detecting memoized-global exec and importlib bypass techniques that industry competitor ModelScan 0.8.8 missed
- Built temporal baseline diffing for rug-pull detection by implementing SafeTensors header inspection, GGUF structural parsing, typosquat detection via Levenshtein distance, and multi-layer obfuscation decode (base64, chr(), string concat), covering 9 distinct attack vectors across the model supply chain
- Reduced model scan bandwidth by 99.9% by using HTTP Range requests to fetch only first 512KB of opcodes and metadata headers up to 16MB, scanning the full GPT-2 repository in 0.5MB instead of downloading 500MB of tensor weights
- Detected 12/12 attack fixtures (CVE-2026-4372, CVE-2026-46432) in 116ms total with 0/5 false positives by supporting 17 file formats, CycloneDX 1.5 SBOM generation, MITRE ATT&CK v19 mapping, and CI templates across 5 platforms

#### **LLM Red-Team Framework** | *Python*

- Built an adversarial prompt-generation and detector-evaluation framework by implementing multi-category attack generation following OWASP LLM Top 10, producing per-category detection-rate reports with structured JSON evidence for guardrail validation

### **PUBLICATION**

#### **A Personalized E-Learning System Using Reinforcement Learning Through Satellite**

IEEE INDICON 2023 | IEEE Xplore: 10440852

### **TECHNICAL SKILLS**

**AI/ML Security:** Agentic AI Security, LLM/RAG Security, Prompt Injection & Jailbreak Testing, MCP/Tool Security, Model Supply-Chain Security, Adversarial ML (FGSM, PGD, C&W), Data Poisoning, Membership Inference, Model Extraction, Differential Privacy

**Standards:** MITRE ATT&CK/ATLAS, OWASP Top 10 for LLM Applications, NIST AI RMF, CycloneDX SBOM

**Cloud & IAM:** AWS IAM, Least-Privilege Enforcement, Trust Relationships, Agent Role Security, Cloud Security, Terraform

**Languages & Tools:** Python, FastAPI, PyTorch, scikit-learn, Docker, Kubernetes, GitHub Actions, CI/CD, SARIF

**Security Eng.:** Threat Modeling, Red Teaming, Guardrails Design, Input/Output Filtering, Access Controls, Audit Logging, Circuit Breakers

### **CERTIFICATIONS & RECOGNITION**

**AWS Academy Graduate:** Cloud Security Foundations | **Technology Innovation Lab:** Honeywell Aerospace & ASU | **KSCST:** Research Grant